

Validity Is Not Difficulty: Supplementary Material

Anonymous CVPR submission

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001 A. SceneIR Schema and Coordinate Conven- 002 tions

003 The released SceneIR records explicit $T_{\text{city} \leftarrow \text{ego}}$,
004 $T_{\text{ego} \leftarrow \text{sensor}}$, and $T_{\text{city} \leftarrow \text{actor}}$ transforms. Physical queries
005 use Actor-local surfaces and known/unknown masks;
006 appearance primitives are referenced only by attachment
007 provenance.

008 B. Frozen Evaluation Protocol

009 The AV2 cohort contains 30 metadata-only selected valida-
010 tion logs. Quantitative logs, qualitative logs, all thresholds,
011 and all category mappings are frozen before method outputs
012 are read. Resource failures remain in the denominator and
013 are reported separately from scientific failures.

014 C. Failure Ledger Template

015 For every failed case we record the failure layer (sensor op-
016 portunity, coordinate/interface, representation, density cal-
017 ibration, or Actor dynamics), the first violated invariant,
018 whether the Actor and hazard event were retained, and the
019 exact non-destructive recovery condition. No failed log may
020 be replaced after quality is read.

021 D. Additional Results

022 Full per-category physical metrics, paired counterfactuals,
023 reliability curves, and 20+ qualitative panels will be inserted
024 after the corresponding frozen milestones.

025 References